Diploma in Computer Science with Artificial Intelligence

Project Report

DCSAI-251-F



School of Computing and Engineering
National Institute of Business Management
Sri Lanka

National Institute of Business Management (NIBM)
Diploma in Computer Science with Artificial Intelligence

Project Report

Web-based vehicle dealership

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Chapter 1

Introduction.

This report documents the creation of a web project known as Elite Autos that is aimed at providing a web-based platform for the automotive sector. The key reason for creating such a web platform is to market cars and services, as well as offer a good means by which people can access information from the website. Essentially, this means that the website is to be used as a virtual car showroom.

The automation industry has experienced a shift in interaction and customer engagement among businesses and customers through digital platforms, and an online presence has become an important factor for the success of automotive businesses. This is because the modern customer wants fast information access, visually engaging information, and the ease of communication through the web.

Additionally, this project allowed for hands-on experience in the application of web development concepts such as layout design, UI design, and frontend development. The report covers the design methodology, system specification, difficulties involved in developing this system, and techniques used to solve these difficulties. Thus, this project of Elite Autos shows how theoretical concepts can be used to solve a real-world web development problem.

Chapter 2

Project objectives.

1. Easy access to information and services,
2. Display available vehicles with pictures and descriptions.
3. Allow potential clients to get in touch or seek services.
4. Offer a contemporary and responsive design that works well on all devices.

Chapter 3

System Features and Requirements

3.1 Functional requirements

User Registration

- The system should allow complainants to register with unique identities.
- Implement mechanisms to prevent multiple registrations per identity.

User Management

- Administrators should be able to manage user accounts and their roles within the system.
- Define user roles (user, administrators) with specific privileges.

Inquiry Submission

- Enable the public to submit inquiries about vehicles

Inquiry Management

- Admin can view all customer inquiries in the data table and change the inquiry status and delete.
- Clicking the "View" button shall open a modal with complete inquiry information and the admin can send availability Email

Vehicle Management

- Admin shall be able to add new vehicles with vehicle details
- Admin shall be able to modify existing vehicle information

Dashboard and Reporting

- Admin dashboard displays key business metrics, shows recent inquiry activity feed and provides quick navigation to management modules

3.2 Nonfunctional requirements

Performance

- Response Time: Page load time shall not exceed 3 seconds on average, and database queries shall execute in under 1 second
- Scalability: System shall support up to 1,000 vehicle records without performance degradation, and the database shall handle up to 10,000 user accounts
- Capacity: Maximum file upload size 5MB per image, and maximum inquiry message length: 5,000 characters

Reliability

- Availability: System uptime target 99% (excluding scheduled maintenance), and scheduled maintenance windows shall be announced in advance
- Data Integrity: All database transactions shall be ACID compliant and foreign key constraints shall maintain referential integrity

- Error Handling: System shall never display raw error messages to users and all errors shall be logged for administrator review

Security

- Authentication: Password minimum length 6 characters. Passwords shall be hashed using the bcrypt algorithm
- Data protection: SQL injection prevention via prepared statements and XSS (Cross-Site Scripting) protection through input sanitization

Usability

- User friendly interface: Intuitive navigation requiring minimal training, and Consistent design patterns across all pages
- Accessibility: Semantic HTML for screen reader compatibility and Sufficient color contrast for readability

Chapter 4

Design

The design for the EliteAutos site aimed to incorporate elements of luxury, professionalism, and Ease of functionality

Header & Navigation:

The header is uniform for all the pages and features the logo of EliteAutos, with links for Home, Inventory, Our Legacy, Concierge, Login, and Register, which helps the visitors navigate through different sections of the site easily.

Homepage :

The home page contains a large hero image. A button labeled “Browse Collection” is used to entice customers to check out the vehicles on display.

Featured Collection:

This section showcases some of the luxury cars. Every card in this section showcases the image, name, and pricing details of the car, along with a button named “View Details”.

Inventory Page:

The "inventory" page provides a carefully selected list of vehicles. The vehicles are shown with their availability status (In Stock / Sold), price, and a "Discover More" option. Filters such as Sedan, SUV, and Sports Cars make browsing easier for consumers.

Legacy Page:

This is the page that conveys the brand history and values of Elite Autos. It contains information regarding the years of experience of the company since 1998, the quality that the company maintains, the excellence of the showroom, and many other stats like the years of experience and the inspection level.

Concierge Page:

In the concierge page, the visitor has the opportunity to access information on contact details, showrooms, working hours, and a form for requesting assistance through the "Request Assistance" option. This is where a visitor can reach the EliteAutos customer service team.

Login and Registration Pages:

The website contains secure Login and registration pages. This is the part of the application where the user enters the system using email and password authentication. The registration form provides an opportunity for new users to register an account using their personal information.

Footer:

The footer is common to all pages and holds information such as company information, quick links, contacts, and copyrights.

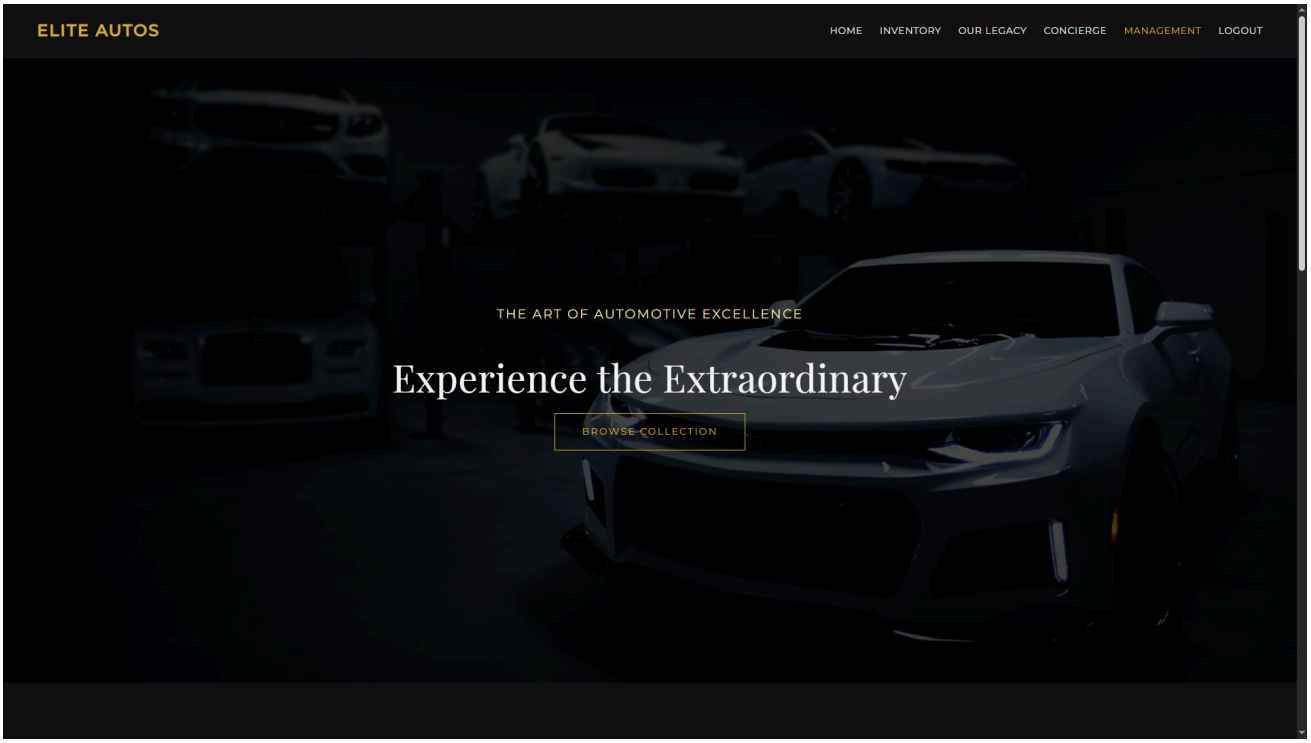
Visual Style

Color Palette:

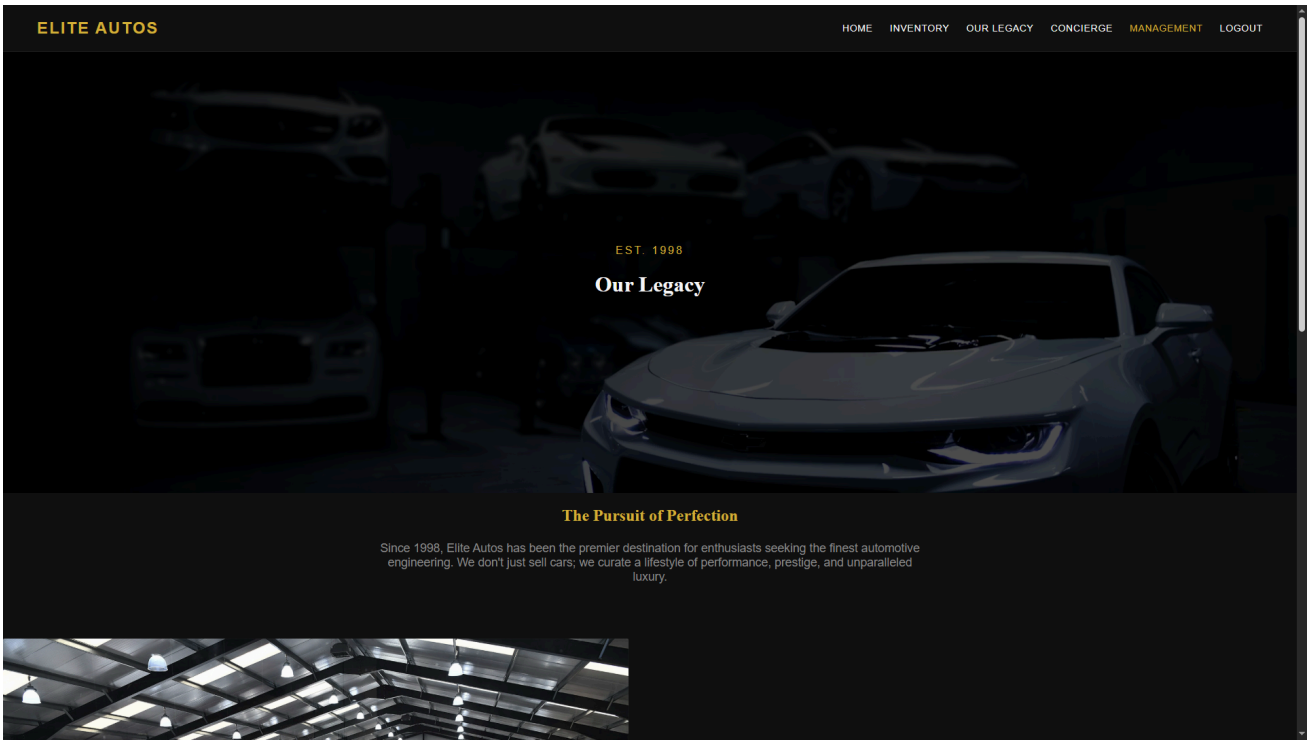
The website has applied dark colors such as black and dark grey to form the background, along with gold and white. These colors make the website look luxurious, elegant, and professional.

Typography:

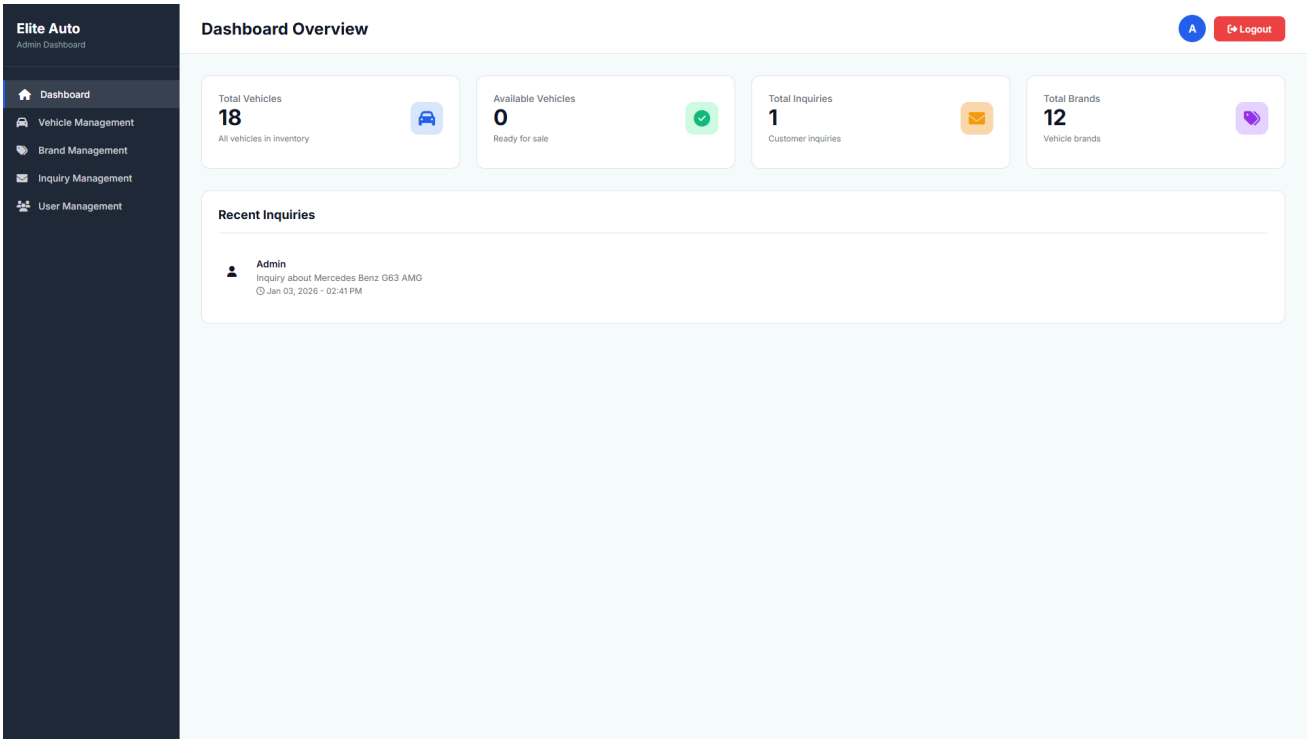
Strong and elegant fonts are adopted for headings with the aim of producing a strong visual effect, with clear and easy-to-read fonts adopted for body text with the aim of facilitating user readability and comfort. Images and Icons: High quality images of luxury cars are used throughout the site to lure visitors and improve engagement



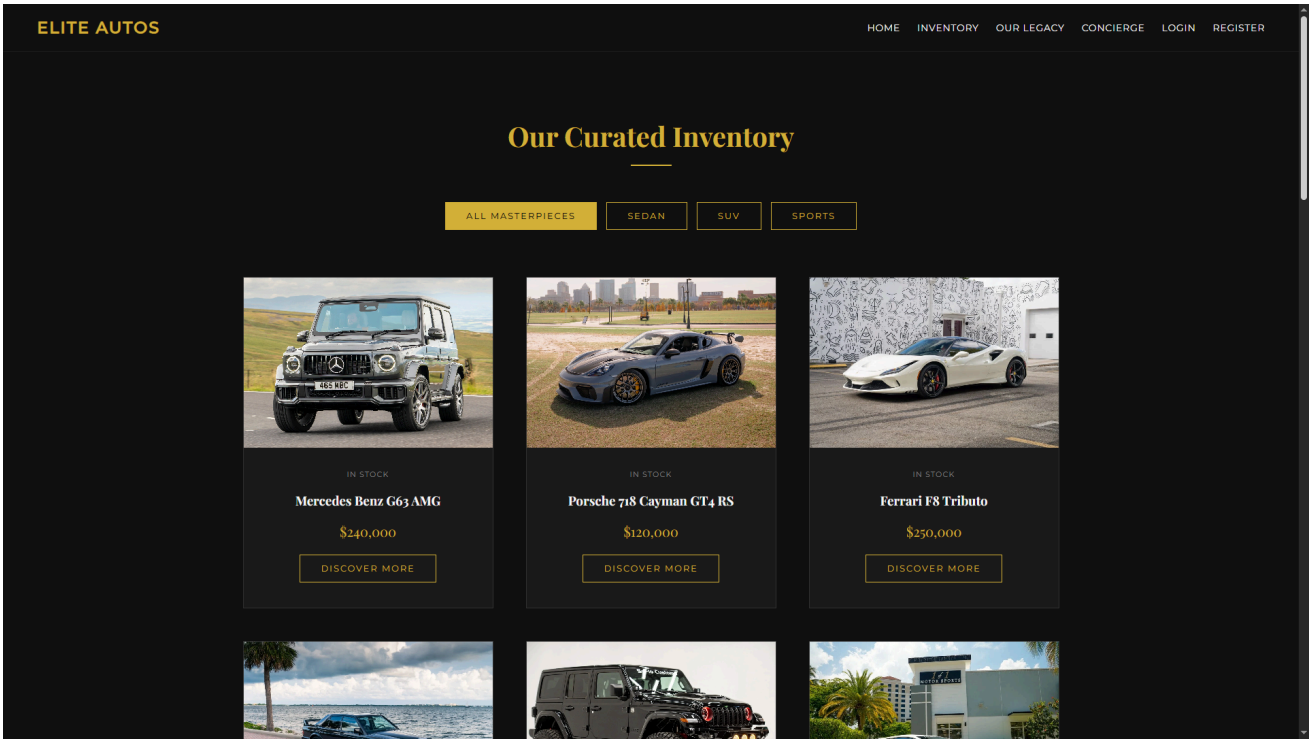
Index page



Our legacy page



Admin dashboard



Inventory page

Chapter 5

Implementation

Front end

HTML:

The markup language helps build the structure for pages such as home, inventory, legacy, concierge, and logging/register page

CSS:

Utilized for application style and design, layout, colors, and responsiveness to give a rich, Professional look.

JavaScript:

Added interactive functionalities, like form validation.

Backend

PHP:

It is used for server-side operations like handling forms, user authentication, vehicle filtering, and managing requests.

It is primarily used

Formspree:

It is used as the backend of the contact form in the Concierge page.

Database

MySQL:

Handles all the required database interactions, such as storing and retrieving car details,

Inquiry services requests for users who are already logged in.

Chapter 6

Challenges

Challenge 1: The website looked Different on Every Browser

The Problem: When I tested the site across different browsers, I found;

- Buttons and layouts looked wrong
- Some features didn't work

The Impact: This was serious because users would have had experience on certain browsers, potentially causing the client to lose business.

Challenge 2: Mobile View Was a Mess

The problem: The site looked great on my laptop, but on phones and tablets:

- The navigation menu collapsed in to and unusable means
- Car images overflowed the screen
- The text was too small to read without zooming in
- The fields were tiny and hard to tap

The impact: This poor experience would have caused many visitors to leave immediately.

Challenge 3: Pages took forever to load

The problem: high-resolution car images caused :

- 8 second load times on mobile

The impact: Users would likely abandon the site before seeing anything

Chapter 7

Solutions implemented

Browser compatibility issues

The website was tested on modern browsers, including Chrome, Firefox, Safari, and Edge. All main functionality works correctly across these browsers with consistent styling and layout. Minor rendering differences were handled through standardized CSS approaches.

Mobile Responsiveness Problems

Added a hamburger-style header for mobile phones

Detected layout issues on smaller screen sizes and mobile devices.

Implemented CSS grid and flexbox to create flexible and responsive layouts.

Redesigned buttons, forms, and navigation elements to be more touch-friendly.

Improved spacing and tap areas for better usability on mobile devices.

Slow Loading Pages

Implemented “object:fit cover” CSS property to ensure the image displays properly without distortions

Set appropriate image dimensions in CSS to prevent layout shifts during loading

Chapter 8

Testing

Manual device testing:

The system was tested on multiple real devices to make sure it works smoothly in different environments. Testing was carried out using a personal iPhone, a Samsung device, and an iPad to confirm that the website is fully responsive and touch-friendly. All core features, such as navigation, vehicle listings, filtering, and forms, were checked on each device. Touch interactions like tapping, scrolling, and swiping worked smoothly without any delays or layout issues.

In addition, desktop testing was performed on both a Windows laptop and Macbook. The website was accessed using different screen sizes to ensure that the layout adjusted properly and all features functioned correctly across operating systems.

Cross Browser Testing:

To ensure consistent behavior across different browsers, the website was tested on the latest versions of Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge. Most browsers displayed the websites correctly without any issues.

During testing, a minor flexbox layout issue was identified and handled through standardized CSS approaches. After these fixes, the website performed consistently on all tested browsers.

User Testing:

User testing was conducted with five friends who are actual car shoppers to gather real-world feedback. They were asked to browse vehicles, navigate through the website, and use the vehicle filtering feature. Their feedback highlighted a few navigation areas that were slightly confusing.

Based on their suggestions, the navigation structure was improved to make it more intuitive, and the vehicle filtering feature was simplified to make it easier to understand. After implementing these improvements, users found the website more user-friendly and easier to navigate.

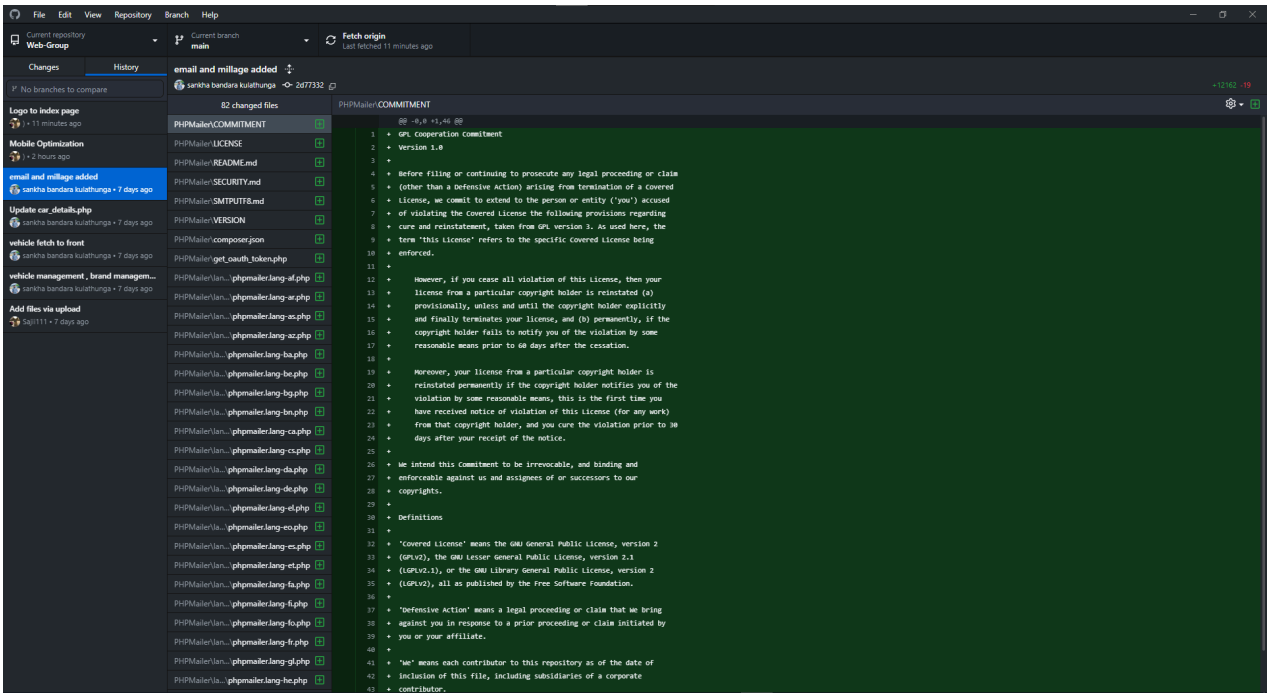
Chapter 9

Collaborative tools

The system development work must be completed within a short period, and it is difficult to share project development work and information with the team members during the study period. The team must come together for project development and communication. Collaborating tools are used to fulfill that requirement. This enables remote coordination with the team members, and effective communication keeps the project development efficient and uniform. This is used to maintain the quality of the development of the system.

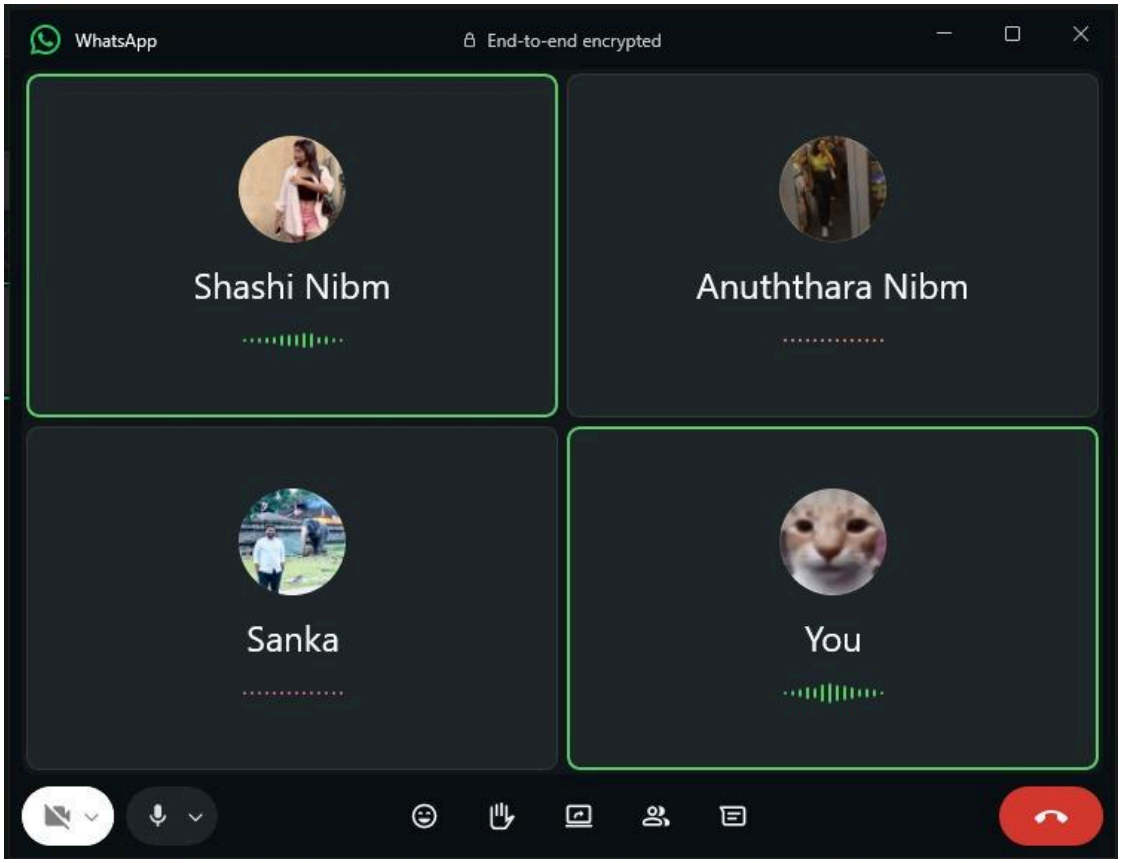
GitHub

GitHub is an open-source hosting tool that helps store and manage code during software development. This facilitates automated and simplified system development. There is a possibility of establishing a main branch related to the project. Team members can create separate profiles. They have provided the space to implement the component of the system related to them in their support. After each component is developed, it is pushed to the master branch. Finally, all the components are integrated into the main branch. Also, commenting on the relevant code will introduce and manage bugs.



WhatsApp

This is a popular social networking platform, and many people use it to communicate with each other. Due to its ease of use, projects were widely used for communication purposes. Team members communicated and discussed the project with the help of text/voice calls and shared documents.



Chapter 10

Conclusion

The Elite Autos project was an incredible learning experience that took us from design concepts to a fully functional, real-world website. We learned that:

- Mobile first isn't just a buzzword. It is essential when 60% of users are on phones.
- Testing early and often saves countless hours of fixing things later.
- Performance optimization can make or break user experience.
- Good project management means planning for the unexpected

The final website not only met all client requirements but also exceeded them, offering faster load times and a better mobile experience than expected. This project solidified our skills in responsive design, performance Optimization, and user centered development.

Website link: https://elitemotors.zociallab.com/